MOS Capacitor Simulator

Equations as implemented in the code

1 Model and notation

The structure is gate – oxide – silicon (p-type or n-type) – grounded bulk contact, in one dimension x.

$$V_t = \frac{kT}{q}, \qquad \phi = \frac{V}{V_t}, \qquad E = -\frac{dV}{dx} \tag{1}$$

The grid nodes are $x_0 \dots x_N$. Node 0 is the gate, node s is the oxide–semiconductor interface and node N is the bulk contact. The oxide mesh is uniform; in the semiconductor the spacing grows geometrically:

$$\Delta x_i = x_{i+1} - x_i, \qquad \Delta x_{s+k} = a\, b^k \quad \Longrightarrow \quad x_{s+k} = t_{ox} + a\,\frac{b^k - 1}{b - 1} \tag{2}$$

Each node owns a control volume. Its semiconductor part has width

$$h_i = \begin{cases} 0 & i < s \quad \text{(oxide)} \\ \tfrac{1}{2}\Delta x_s & i = s \quad \text{(interface: half a cell)} \\ \tfrac{1}{2}\left(\Delta x_{i-1} + \Delta x_i\right) & s < i < N \end{cases} \tag{3}$$

Differences of the normalised potential between neighbouring nodes are written $\Delta\phi_i = \phi_{i+1} - \phi_i$, and n_{int} is the intrinsic carrier concentration.

2 Current densities (Scharfetter–Gummel)

$$J_n = q\mu_n n E + qD_n\frac{dn}{dx} = kT\mu_n\left(-n\frac{d\phi}{dx} + \frac{dn}{dx}\right) \tag{4}$$

$$J_p = q\mu_p p E - qD_p\frac{dp}{dx} = -kT\mu_p\left(p\frac{d\phi}{dx} + \frac{dp}{dx}\right) \tag{5}$$

The current is a flux between two nodes, so it is defined on the element $i+\frac{1}{2}$ between nodes i and $i{+}1$. Taking the potential linear inside an element and writing $n = u\,e^{\phi}$ gives $J_n = kT\mu_n e^{\phi}\,du/dx$. Integrating $J_n e^{-\phi}$ over the element with constant J_n:

$$J_{n,i+\frac{1}{2}} = \frac{kT\mu_n}{\Delta x_i}\Big[B(\Delta\phi_i)\,n_{i+1} - B(-\Delta\phi_i)\,n_i\Big] \tag{6}$$

with the Bernoulli function

$$B(x) = \frac{x}{e^x - 1}, \qquad B(x)\,e^x = B(-x) \tag{7}$$

For holes take $p = u\,e^{-\phi}$, so $J_p = -kT\mu_p e^{-\phi}\,du/dx$ and

$$J_{p,i+\frac{1}{2}} = \frac{kT\mu_p}{\Delta x_i}\Big[B(\Delta\phi_i)\,p_i - B(-\Delta\phi_i)\,p_{i+1}\Big] \tag{8}$$

2.1 Quasi-Fermi form

With the quasi-Fermi potentials Φ_n, Φ_p defined by

$$n = n_{int} \exp\!\left(\frac{V - \Phi_n}{V_t}\right), \qquad p = n_{int} \exp\!\left(\frac{\Phi_p - V}{V_t}\right) \tag{9}$$

the currents (6), (8) become, exactly,

$$\frac{J_{n,i+\frac{1}{2}}}{q} = -\frac{V_t \mu_n}{\Delta x_i} B(\Delta\phi_i)\, n_{i+1} \left(e^{d_n} - 1\right), \qquad d_n = \frac{\Phi_{n,i+1} - \Phi_{n,i}}{V_t} \tag{10}$$

$$\frac{J_{p,i+\frac{1}{2}}}{q} = -\frac{V_t \mu_p}{\Delta x_i} B(\Delta\phi_i)\, p_i \left(e^{d_p} - 1\right), \qquad d_p = \frac{\Phi_{p,i+1} - \Phi_{p,i}}{V_t} \tag{11}$$

In this form drift and diffusion are never subtracted from each other, so a small net current is computed accurately. A current is zero exactly when the quasi-Fermi potential is flat. This form is used in the small-signal and transient solvers.

3 Steady state (equilibrium and DC bias)

3.1 Poisson's equation

$$\frac{d}{dx}\left(\varepsilon \frac{dV}{dx}\right) = -\rho, \qquad \rho = q\,(N_D - N_A + p - n) \tag{12}$$

Integrating over the control volume of node i (there is no charge in the oxide, where $h_i = 0$):

$$F_i \equiv \varepsilon_{i+\frac{1}{2}} \frac{V_{i+1} - V_i}{\Delta x_i} - \varepsilon_{i-\frac{1}{2}} \frac{V_i - V_{i-1}}{\Delta x_{i-1}} + q\, h_i\, (N_D - N_A + p_i - n_i) + Q_f\, \delta_{i,s} = 0 \tag{13}$$

where $\varepsilon_{i+\frac{1}{2}} = \varepsilon_{ox}$ for $i < s$ and ε_{si} for $i \geq s$. At the interface node $i = s$ the two permittivities differ, which enforces continuity of εE; the fixed oxide charge Q_f (per unit area) sits at this node only.

Boundary conditions. V is measured from the bulk, so $V_N = 0$. The gate and the semiconductor have different work functions, so a gate voltage V_G puts the gate node at

$$V_0 = V_G - \phi_{ms}, \qquad \phi_{ms} = \phi_m - \phi_s, \qquad \phi_s = \chi + \frac{E_g}{2q} - V_t \ln \frac{n_{bulk}}{n_{int}} \tag{14}$$

with ϕ_m the gate work function, χ the electron affinity and n_{bulk} the electron concentration at the bulk contact. This covers both substrate types: $N_A > N_D$ is p-type ($n_{bulk} < n_{int}$) and $N_D > N_A$ is n-type. The bands are flat ($V = 0$ everywhere) at the flat-band voltage

$$V_{FB} = \phi_{ms} - \frac{Q_f}{C_{ox}}, \qquad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \tag{15}$$

Equilibrium is $V_G = 0$, which is in general not the flat-band condition.

The equation is non-linear because n and p depend exponentially on V. One Newton step $J\,\Delta V = -F$ is taken with the carriers following the potential at fixed quasi-Fermi potentials:

$$J_{i,i+1} = \frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x_i}, \qquad J_{i,i-1} = \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x_{i-1}}, \qquad J_{i,i} = -\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x_i} - \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x_{i-1}} - \frac{q\, h_i\, (n_i + p_i)}{V_t} \tag{16}$$

followed by $V \leftarrow V + \Delta V$, $n \leftarrow n\, e^{\Delta V/V_t}$, $p \leftarrow p\, e^{-\Delta V/V_t}$. Steps larger than V_t are damped to $\pm V_t \ln(1 + |\Delta V|/V_t)$.

3.2 Carrier continuity

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - R, \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - R \tag{17}$$

Recombination is Shockley–Read–Hall:

$$R = \frac{np - n_{int}^2}{S}, \qquad S = \tau_p\,(n + n_{int}) + \tau_n\,(p + n_{int}) \tag{18}$$

In steady state, integrating over the control volume of node i:

$$\frac{1}{q}\left(J_{n,i+\frac{1}{2}} - J_{n,i-\frac{1}{2}}\right) = h_i R_i, \qquad -\frac{1}{q}\left(J_{p,i+\frac{1}{2}} - J_{p,i-\frac{1}{2}}\right) = h_i R_i \tag{19}$$

Inserting (6), (8) and dividing by $V_t\mu$, with S_i and the other carrier taken from the previous iteration:

$$\alpha_{i-1}n_{i-1} + \alpha_i n_i + \alpha_{i+1}n_{i+1} = -\frac{h_i\,n_{int}^2}{V_t\mu_n S_i} \tag{20}$$

$$\alpha_{i+1} = \frac{B(\Delta\phi_i)}{\Delta x_i}, \quad \alpha_{i-1} = \frac{B(-\Delta\phi_{i-1})}{\Delta x_{i-1}}, \quad \alpha_i = -\left[\frac{B(-\Delta\phi_i)}{\Delta x_i} + \frac{B(\Delta\phi_{i-1})}{\Delta x_{i-1}}\right] - \frac{h_i\,p_i}{V_t\mu_n S_i} \tag{21}$$

$$\beta_{i-1}p_{i-1} + \beta_i p_i + \beta_{i+1}p_{i+1} = -\frac{h_i\,n_{int}^2}{V_t\mu_p S_i} \tag{22}$$

$$\beta_{i+1} = \frac{B(-\Delta\phi_i)}{\Delta x_i}, \quad \beta_{i-1} = \frac{B(\Delta\phi_{i-1})}{\Delta x_{i-1}}, \quad \beta_i = -\left[\frac{B(\Delta\phi_i)}{\Delta x_i} + \frac{B(-\Delta\phi_{i-1})}{\Delta x_{i-1}}\right] - \frac{h_i\,n_i}{V_t\mu_p S_i} \tag{23}$$

Boundary conditions:

- Interface ($i = s$): no current enters the oxide, $J_{s-\frac{1}{2}} = 0$. The equations above are used with the $i-1$ terms dropped and $h_s = \frac{1}{2}\Delta x_s$.
- Bulk contact ($i = N$): n_N, p_N keep the charge-neutral equilibrium values, from $p - n = N_A - N_D$ and $np = n_{int}^2$.

Poisson's equation and the two continuity equations are solved one after the other and repeated until V, n and p stop changing. The gate voltage is moved in steps of at most 0.1 V, each step starting from the previous solution.

3.3 Solving the continuity equations accurately

Both equations have the form $\mathcal{F}_i - \mathcal{F}_{i-1} - \sigma_i y_i = -g_i$ with flux $\mathcal{F}_i = u_i y_{i+1} - d_i y_i$ and $u_i, d_i, \sigma_i, g_i \geq 0$. The fluxes are up to 10^{17} times larger than the recombination terms, so ordinary Gaussian elimination loses them to round-off. The elimination is therefore arranged so that it never subtracts: after eliminating nodes $s \ldots i$, node i obeys $u_i y_{i+1} - (d_i + e_i)\,y_i = -b_i$ with

$$e_s = \sigma_s, \quad b_s = g_s, \qquad e_{i+1} = \sigma_{i+1} + \frac{u_i\,e_i}{d_i + e_i}, \qquad b_{i+1} = g_{i+1} + \frac{d_i\,b_i}{d_i + e_i} \tag{24}$$

and the back substitution is $y_i = (u_i y_{i+1} + b_i)\,/\,(d_i + e_i)$.

3.4 Deep depletion

The steady state always contains the inversion layer. If the gate is swept from accumulation towards inversion faster than minority carriers can be generated, no inversion layer forms and the depletion region keeps widening. This fast-sweep limit is modelled by freezing the minority carrier:

- the sweep starts from the steady state at the accumulation end of the gate-voltage range;
- the minority carrier (electrons for p-type, holes for n-type) keeps the profile it has there. It is not solved from the continuity equation and not rescaled in the Newton step, so it also drops out of the Jacobian: $J_{i,i}$ contains only the majority carrier;
- the majority carrier and Poisson's equation are solved as before.

The capacitance equals the high-frequency one up to threshold and keeps falling beyond it, as $C = (1/C_{ox} + W/\varepsilon_{si})^{-1}$ with W the depletion width.

4 Small-signal AC analysis

A small sinusoidal signal is added to the DC gate bias, and every quantity is written as its DC value (subscript 0) plus a phasor:

$$V_G = V_{DC} + \tilde{v}\, e^{j\omega t}, \qquad V = V_0 + \tilde{V} e^{j\omega t}, \qquad \Phi_n = \Phi_{n0} + \tilde{W}_n e^{j\omega t}, \qquad \Phi_p = \Phi_{p0} + \tilde{W}_p e^{j\omega t} \tag{25}$$

Only terms linear in the perturbation are kept, and $\partial/\partial t \to j\omega$. From (9) the carrier phasors are

$$\tilde{n} = \frac{n_0}{V_t}\left(\tilde{V} - \tilde{W}_n\right), \qquad \tilde{p} = \frac{p_0}{V_t}\left(\tilde{W}_p - \tilde{V}\right) \tag{26}$$

4.1 Linearised equations

No DC current flows in a MOS capacitor, so the DC quasi-Fermi potentials are flat ($d_n = d_p = 0$) and $B(\Delta\phi_{0,i})\, n_{0,i+1} = B(-\Delta\phi_{0,i})\, n_{0,i} \equiv \bar{n}_i$, $B(\Delta\phi_{0,i})\, p_{0,i} = B(-\Delta\phi_{0,i})\, p_{0,i+1} \equiv \bar{p}_i$. Linearising (10), (11) then leaves plain conductances:

$$\frac{\tilde{J}_{n,i+\frac{1}{2}}}{q} = -\frac{\mu_n \bar{n}_i}{\Delta x_i}\left(\tilde{W}_{n,i+1} - \tilde{W}_{n,i}\right), \qquad \frac{\tilde{J}_{p,i+\frac{1}{2}}}{q} = -\frac{\mu_p \bar{p}_i}{\Delta x_i}\left(\tilde{W}_{p,i+1} - \tilde{W}_{p,i}\right) \tag{27}$$

Linearising (18) around the DC solution (R_0, S_0 are the DC values):

$$\tilde{R} = r_n \tilde{n} + r_p \tilde{p}, \qquad r_n = \frac{p_0 - R_0\tau_p}{S_0}, \qquad r_p = \frac{n_0 - R_0\tau_n}{S_0} \tag{28}$$

The three equations of node i are the linearised (13) and (17):

$$\varepsilon_{i+\frac{1}{2}} \frac{\tilde{V}_{i+1} - \tilde{V}_i}{\Delta x_i} - \varepsilon_{i-\frac{1}{2}} \frac{\tilde{V}_i - \tilde{V}_{i-1}}{\Delta x_{i-1}} + q\, h_i\, (\tilde{p}_i - \tilde{n}_i) = 0 \tag{29}$$

$$-\frac{1}{q}\left(\tilde{J}_{n,i+\frac{1}{2}} - \tilde{J}_{n,i-\frac{1}{2}}\right) + h_i\left(\tilde{R}_i + j\omega\, \tilde{n}_i\right) = 0 \tag{30}$$

$$-\frac{1}{q}\left(\tilde{J}_{p,i+\frac{1}{2}} - \tilde{J}_{p,i-\frac{1}{2}}\right) - h_i\left(\tilde{R}_i + j\omega\, \tilde{p}_i\right) = 0 \tag{31}$$

ϕ_{ms} and Q_f are constants, so they do not appear.

4.2 Boundary conditions

- Gate: $\tilde{V}_0 = \tilde{v}$. The oxide has no charge, so $\tilde{V}$ is linear in it and the oxide acts as the capacitor C_{ox} between gate and interface.
- Interface ($i = s$): $\tilde{J}_{n,s-\frac{1}{2}} = \tilde{J}_{p,s-\frac{1}{2}} = 0$, with $h_s = \frac{1}{2}\Delta x_s$.
- Bulk contact ($i = N$): $\tilde{V}_N = \tilde{W}_{n,N} = \tilde{W}_{p,N} = 0$.

4.3 Solution

The AC problem is linear, so it is solved once per bias point and frequency, with no iteration. With $X_i = (\tilde{V}_i, \tilde{W}_{n,i}, \tilde{W}_{p,i})^T$, the three equations of node i are

$$(U_i X_{i+1} - D_i X_i) - (U_{i-1} X_i - D_{i-1} X_{i-1}) - L_i X_i = -g_i \tag{32}$$

with 3×3 complex matrices: $U_i = D_i = \mathrm{diag}(\varepsilon_{si},\ \mu_n \bar{n}_i,\ \mu_p \bar{p}_i)\,/\Delta x_i$ hold the flux coefficients of element $i+\frac{1}{2}$, L_i the local terms (charge, recombination, $j\omega$), and g_i the sources. The oxide enters at the interface node only: C_{ox} is added to the first diagonal entry of L_s, and $g_s = (C_{ox}\tilde{v}, 0, 0)^T$; all other $g_i = 0$. This is the block version of Section 3.3: the fluxes are never subtracted from each other. After eliminating nodes $s \ldots i$, node i obeys $U_i X_{i+1} - (D_i + E_i) X_i = -b_i$ with

$$E_s = L_s, \qquad b_s = g_s \tag{33}$$

$$E_{i+1} = L_{i+1} + E_i\,(D_i + E_i)^{-1} U_i, \qquad b_{i+1} = g_{i+1} + D_i\,(D_i + E_i)^{-1} b_i \tag{34}$$

and back substitution $X_i = (D_i + E_i)^{-1}\,(U_i X_{i+1} + b_i)$ starting from $X_N = 0$.

4.4 Capacitance and conductance

The gate charge per unit area is $Q = \varepsilon_{ox} E_{ox}$, so its phasor is $\tilde{Q} = C_{ox}(\tilde{v} - \tilde{V}_s)$ and

$$\tilde{C} = \frac{\tilde{Q}}{\tilde{v}} = C_{ox}\left(1 - \frac{\tilde{V}_s}{\tilde{v}}\right), \qquad C = \mathrm{Re}\,\tilde{C}, \qquad \frac{G}{\omega} = -\mathrm{Im}\,\tilde{C} \tag{35}$$

C is the capacitance and G the conductance per unit area at angular frequency ω (the admittance is $j\omega\tilde{C}$). At low frequency the inversion layer follows the signal and C returns to about C_{ox} in inversion; at high frequency it cannot, and C stays at its minimum. G/ω peaks at the frequency where the change-over happens.

Two kinds of sweep are done: a bias sweep (C and G/ω versus V_G at a few frequencies) and a frequency sweep (C and G/ω versus f at a few gate voltages).

5 Transient simulation

The steady state and the frozen-carrier deep depletion are the two limits of a gate that moves infinitely slowly or infinitely fast. In between, the equations are solved in time.

5.1 Time stepping

The unknowns at each node are $X_i = (V_i, \Phi_{n,i}, \Phi_{p,i})$, with the currents in the form (10), (11) and $np - n_{int}^2 = n_{int}^2\left(e^{(\Phi_p - \Phi_n)/V_t} - 1\right)$ in (18). Backward Euler from time t to $t + \Delta t$, with all terms

at the new time except n^{old}, p^{old}:

$$\varepsilon_{i+\frac{1}{2}} \frac{V_{i+1} - V_i}{\Delta x_i} - \varepsilon_{i-\frac{1}{2}} \frac{V_i - V_{i-1}}{\Delta x_{i-1}} + q\, h_i \left(N_D - N_A + p_i - n_i\right) + Q_f\, \delta_{i,s} = 0 \tag{36}$$

$$-\frac{1}{q}\left(J_{n,i+\frac{1}{2}} - J_{n,i-\frac{1}{2}}\right) + h_i \left(R_i + \frac{n_i - n_i^{old}}{\Delta t}\right) = 0 \tag{37}$$

$$-\frac{1}{q}\left(J_{p,i+\frac{1}{2}} - J_{p,i-\frac{1}{2}}\right) - h_i \left(R_i + \frac{p_i - p_i^{old}}{\Delta t}\right) = 0 \tag{38}$$

The boundary conditions are those of the steady state, with the gate node at $V_G(t + \Delta t) - \phi_{ms}$. The three equations are solved together by Newton's method. The linear system of each Newton iteration has the block form of Section 4.3, with U_i, D_i, L_i the derivatives of the fluxes and local terms with respect to X and the residual as g_i, and is solved by the same elimination. Updates larger than $5V_t$ are clipped.

5.2 Simulations

Both start from the steady state at the accumulation end of the gate-voltage range. The capacitance reported is the small-signal one of Section 4 at the highest frequency, around the state at each time.

- Sweep-rate dependent C–V: the gate is ramped towards inversion at a rate dV_G/dt, with $\Delta t = \Delta V_G/(dV_G/dt)$ per step. Slow ramps follow the equilibrium high-frequency curve and fast ramps the deep-depletion curve. The change-over is where the charging current $C_{ox}\, dV_G/dt$ equals the generation current, about $q\, n_{int} W/(2\tau)$.
- C–t: the gate is stepped into inversion in a very short time and then held. The device starts in deep depletion and the capacitance recovers to the equilibrium high-frequency value as generation builds up the inversion layer. The time steps grow geometrically after the step.